

Capstone limitations and revision

Project: Image-pattern Network Investigation

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Capstone controlled experiments

Without opening the old lesson, explain one key idea from Capstone controlled experiments. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Documenting a network experiment

Without opening the old lesson, explain one key idea from Documenting a network experiment. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Uncertainty and failure analysis

Without opening the old lesson, explain one key idea from Uncertainty and failure analysis. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Compare with the baseline for Capstone limitations and revision. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Capstone limitations and revision. Proposed explanation: “Passing a criterion means no failures remain”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Evaluate new evidence independently

A frozen ReLU candidate with $a = 3$ is compared with baseline $a = 1$ on final rows (2,6),(4,12),(-1,-3),(-2,-6). Compute both MSE values and candidate MSE on negative inputs. Write a bounded conclusion.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Trace final predictions

For F1–F4 calculate candidate residuals and squared errors.

My reasoning and evidence

6. W2 Compare with the baseline

Compute both overall MSE values and the candidate negative-input MSE. Apply the strict final criterion.

My reasoning and evidence

7. W3 Explain the architecture limit

Why cannot extra training alone make this ReLU output negative? Name a proposed change and the evidence it would require.

My reasoning and evidence

8. W4 Evaluate new evidence independently

A frozen ReLU candidate with $a = 3$ is compared with baseline $a = 1$ on final rows (2,6),(4,12),(-1,-3),(-2,-6). Compute both MSE values and candidate MSE on negative inputs. Write a bounded conclusion.

My reasoning and evidence

9. W5 Label a revision honestly

After inspecting F1–F4, a student changes to prediction $2 \cdot x$ and reports “new final MSE 0”. Explain the accurate label and the next evaluation step.

My reasoning and evidence

10. Project checkpoint and reflection

For Image-pattern Network Investigation, use this week’s independent task as a rehearsal for the final evidence pack: A frozen ReLU candidate with $a = 3$ is compared with baseline $a = 1$ on final rows (2,6), (4,12), (-1,-3), (-2,-6). Compute both MSE values and candidate MSE on negative inputs. Write a bounded conclusion. Keep actual observations separate from expected results, preserve failures, compare a baseline and explain the limits.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Capstone limitations and revision

Revise a limited claim after a slice failure. A toy network works better on centred patterns than on shifted patterns, so the team narrows its initial claim.

Centred slice: 7/8 correct. Shifted slice: 3/8 correct. The network and label rubric are frozen.

AI ACADEMY / AI-5/35

Toy network condition slices

correct cases out of 8 per condition



Bar length shows the supplied value; the marked baseline is zero.

Capstone limitations and revision

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Compute slice rates

Centred is 87.5%; shifted is 37.5%.



2. Revise the claim

Do not describe the network as shift-robust on this evidence.



3. Plan a fair revision

Use separate shifted development patterns, then assess the frozen revision on reserved shifted cases.

CHANGE ONE THING

Train the revision on all eight failed-slice evaluation patterns and rescore them.

Explain the changed result using the matching lesson.

Calculate the slice rates and explain the claim revision. If the team trains on the shifted evaluation patterns, what kind of evidence does their rerun provide?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For the Image-pattern Network Investigation, save the labelled trace and the changed-case explanation beside the experiment record. State exactly what this example supports before linking it to the wider capstone claim.

13. Project portfolio milestone

Project: Image-pattern Network Investigation

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with evidence.

For the Image-pattern Network Investigation, save the labelled trace and the changed-case explanation beside the experiment record. State exactly what this example supports before linking it to the wider capstone claim.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Revise a limited claim after a slice failure. A toy network works better on centred patterns than on shifted patterns, so the team narrows its initial claim.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Use separate shifted development patterns, then assess the frozen revision on reserved shifted cases.

1. Compute slice rates
2. Revise the claim
3. Plan a fair revision

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Plan a fair revision Use separate shifted development patterns, then assess the frozen revision on reserved shifted cases. A limitation guides a better next experiment. Training on a failed evaluation example may help development, but that exposed example cannot remain the untouched evidence for the revision.

Practice 2: explain the deciding evidence

Changed-case practice: Train the revision on all eight failed-slice evaluation patterns and rescore them. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. Treat the score as exposed-case development, not independent evidence of shift robustness.
3. This one example guarantees correct results for every future input.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Treat the score as exposed-case development, not independent evidence of shift robustness. The evaluation records now influenced learning and model selection.

Repair a missing building block

Review Capstone controlled experiments (ai-5/34). Goal: Trace matched two-update learning-rate experiments. Short practice: Compute the two correct counts and identify the changed case. Why would changing B's training data weaken the conclusion about its added connection?. Return to this week and explain what you changed.

Review Documenting a network experiment (ai-5/32). Goal: Record a network experiment with enough detail to replay its calculations. Short practice: What additional evidence is needed for a final performance claim? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Uncertainty and failure analysis (ai-5/29). Goal: Distinguish a predicted class, confidence score and observed correctness. Short practice: What is C's positive-class probability and chosen-label confidence? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

